

Recall from last class:

- Let $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$ be an m by n matrix. We express A by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

where a_{ij} is the entry in the i -th row and j -th column.

- **Theorem:** Let A , B , and C be matrices of the same size, and let r and s be real numbers. Then the following properties hold:

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.
3. $A + 0 = A$.
4. $r(A + B) = rA + rB$.
5. $(r + s)A = rA + sA$.
6. $r(sA) = rsA$.

- If A is a $m \times n$ matrix and B is an $n \times p$ matrix with columns $\vec{b}_1, \dots, \vec{b}_p$, then AB is the $m \times p$ matrix defined by

$$AB = A[\vec{b}_1 \ \dots \ \vec{b}_p] = [A\vec{b}_1 \ \dots \ A\vec{b}_p].$$

- Note

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}.$$

- **Theorem:**

1. $A(BC) = (AB)C$.
2. $A(B + C) = AB + AC$.
3. $(B + C)A = BA + CA$.
4. $r(AB) = (rA)B = A(rB)$.
5. $I_m A = A = A I_n$.

- Given an $m \times n$ matrix A , then the **transpose** of A , denoted A^T is the $n \times m$ matrix given by taking the columns of A as rows.

- **Theorem:** Let A be an m by n matrix, B be an n by p matrix, and $r \in \mathbb{R}$.

1. $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$.
3. $(rA)^T = rA^T$.
4. $(AB)^T = B^T A^T$.

Definition 1. An n by n matrix A is **invertible** if there is an n by n matrix C so that $AC = I_n$ and $CA = I_n$. We denote the inverse of A by A^{-1} .

Natural Questions:

1. Do inverses always exist?
2. If not, when do they exist?
3. If they exist, are they unique?
4. How can we find them?

1. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Determine when A is invertible and find A^{-1} .

Theorem: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Definition 2. We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

Theorem: If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.

2. Prove the Theorem.

Theorem: Suppose A and B are n by n invertible matrices, then the following hold.

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) $(AB)^{-1} = B^{-1}A^{-1}$.

(c) $(A^T)^{-1} = (A^{-1})^T$.

3. Prove the Theorem.

Definition 3. An elementary matrix E is a matrix obtained from the identity matrix by a single elementary row operation.

4. Give some examples of elementary matrices.

Theorem: A is invertible if and only if A is row equivalent to the identity matrix. Furthermore, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

5. Prove the theorem.

6.

(a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. Find A^{-1} if possible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. Find A^{-1} if possible.

Invertible Matrix Theorem Let A be an $n \times n$ matrix, then the following are all equivalent.

- (a) A is invertible.
- (b) A is row-equivalent to I_n .
- (c) A has n pivot positions.
- (d) $A\vec{x} = \vec{0}$ has only the trivial solution.
- (e) The columns of A are linearly independent.
- (f) $x \mapsto A\vec{x}$ is one-to-one (injective).
- (g) For all $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) $\vec{x} \mapsto A\vec{x}$ is onto (surjective).
- (j) There exists a matrix C such that $CA = I_n$.
- (k) There exists a matrix D such that $AD = I_n$.
- (l) A^T is invertible.

7. Prove the theorem.

8. Determine if the following matrices are invertible.

(a) Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & -4 & 0 \\ 8 & 5 & -3 \end{bmatrix}$.

(b) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let A be the associated matrix. Then T is invertible if and only if A is invertible.

9. Prove the theorem.